

HOW ACCURATE IS GOOD ENOUGH?

Approximating a Civil Engineering Function Using Infinite Series — Written Recommendation (Deliverable 5)

[1. Numerical Tables](#) [2. Written Recommendation](#)

2. Written Recommendation

Engineering Recommendation

Use the **Maclaurin series with 4 terms** as the general-purpose approximation for computing $y = 20 \cdot \sin(\theta)$ for angles from 0° to 30° .

1. Number of Terms Required

- Angles $< 5^\circ$: 2 terms suffice (error $< 0.001\%$)
- Angles 5° to 15° : 3 terms suffice (error $< 0.001\%$)
- Angles 15° to 30° : 4 terms needed (error $< 0.01\%$)
- 4 terms achieve $< 0.1\%$ error for *all* angles up to 30°

2. Percentage Error Achieved

- With 3 terms at 30° : error $\approx 0.0004\%$ (well within the 0.1% tolerance)
- With 4 terms at 30° : error $\approx 0.000002\%$

3. Convergence Behavior

- The Maclaurin series converges factorially — extremely fast.
- Each added term reduces the error by roughly a factor of $\theta^2/[(2k)(2k+1)]$, i.e. the error drops by orders of magnitude per term.
- At 30° the 1-term error is 4.72% ; the 4-term error is effectively zero.

4. Computational Simplicity vs Accuracy

- Maclaurin requires only θ ; no expansion point needs to be chosen.
- Implementation uses only addition, multiplication and division.
- 4 terms is computationally trivial compared with a full sine evaluation.
- Taylor centered at 10° is only marginally better near 10° and is much worse for small angles; it also requires the center point and its derivatives.

5. Valid Angle Range

- With 4 terms: accurate for all angles from 0° to 30° (0.1% tolerance).
- With 3 terms: accurate for angles from 0° to roughly 20° .

Conclusion

For a civil engineer computing vertical components $y = L \cdot \sin(\theta)$ over the usual surveying/structural range (0° – 30°), the **Maclaurin series with 4 terms** is the best compromise:

- Error well below the 0.1% tolerance for the entire range,
- simple to implement and verify,
- fast, stable convergence with negligible computational cost,
- no sensitivity to the choice of an expansion center.

The Taylor series centered at 10° would only be preferred if all angles of interest clustered tightly around 10° . For a general-purpose engineering tool, the Maclaurin series is simpler and equally effective — and for angles below about 4° , even a single term ($\sin(\theta) \approx \theta$) is sufficient.